



Global Estimation of the Soil Erosion Cover-Management Factor for Non-Agricultural Land ($C_{non-crop}$) Using Google Earth Engine

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The cover-management factor for non-agricultural land ($C_{non-crop}$) is a critical parameter in the USLE-based soil erosion models, and its estimation accuracy directly affects the reliability of regional and global soil erosion simulations. This study developed a dynamic global estimation framework for $C_{non-crop}$ by integrating a pixel-based spectral mixture model (NDVI-SWIR32) and rainfall erosivity weighting within the Google Earth Engine platform. Using MOD43A4 and MOD09GA data, we retrieved global fractional cover of photosynthesizing vegetation (f_{PV}), non-photosynthesizing vegetation (f_{NPV}), and bare soil (f_{BS}) at 500 m resolution from 2000 to 2024. The f_{NPV} during the non-growing season in deciduous forests was explicitly incorporated as a proxy for annual understory coverage to better represent its role in erosion mitigation. Combined with ERA5 rainfall data and land use information, soil loss ratios across 24 half-month periods were weighted by rainfall erosivity to generate a continuous global $C_{non-crop}$ product. The results show that the mean $C_{non-crop}$ values for forests, shrublands, and grasslands were 0.017, 0.014, and 0.029, respectively, which align closely with values reported in the literature (0.020, 0.018, and 0.032), with absolute errors ranging between 0.003 and 0.004. This confirms the stability and applicability of the proposed method across different ecological zones. Approximately 82.8% of the global land area exhibited a $C_{non-crop}$ value below 0.2, primarily distributed in temperate and subtropical regions. High $C_{non-crop}$ values (>0.5) were concentrated in arid and semi-arid regions such as North Africa and Central Asia, indicating weaker vegetation protection and higher ecosystem vulnerability. The explicit inclusion of understory vegetation significantly improved the parameterization of soil erosion factors in forest ecosystems and effectively reduced the systematic bias associated with relying solely on canopy cover.